Training the model.

i) Spliting the data into Feature and Target.
ii) Normalize / Scale the Features.
iii) Split into Train and Test sets.
iv) Train the Model (For Example : Linear Regression.)
v) Make Predictions.
vi) Evaluate the Model
vii) Optional - Visualize Results.

Features.

* Also called as
  Input,
  Predictor,
  Independent variable.
* A Feature is an input variable that the model uses to make prediction.
* It's the "know" data.
* comes from the column in your dataset (expect the target).

The Model learns patterns from these features.

Target.

* Also called as
  output,
  Label,
  dependent.

* "unknown" value your model is trying to estimate based on the input features.
* In supervised learning, we train the model using known features and known target values.

Features = The data that we given.
Target = What we are gonna predict.

Train and Test Data sets.

Training Data.

* This is the portion of the dataset used to train your model.
* The model learns patterns from this data - how the features relate to the target.
* Usually around 70-80% of full dataset.

Testing Data.

* This is the portion of data kept aside to evaluate the model's performance.
* The model has never seen this data during training.
* Help us understand how the model performs on unseen/real-world data.
* Usually around 20-30% of the full dataset.

Why do we split the Data.

* If we train and test the model on the same data, the model may simply memorize the data (overfitting), giving misleading high accuracy.

i) The model learns from the training dataset.
ii) the model is tested on new data (test set) to check how well it generalizes.

random_state

Like seeding a random number generator.
It ensures that the random splitting of data is the same every time you run the code.